

Exercise J01

Q1. In a triangle ABC, right angled at C, $\tan A + \tan B$ is equal to

- (A) $a + b$ (B) $\frac{c^2}{ab}$ (C) $\frac{a^2}{bc}$ (D) $\frac{b^2}{ac}$

Ans. B

Sol. $\angle C = 90^\circ$ $A+B = 90$

$$\begin{aligned}\tan A + \tan B &= \frac{\sin A}{\cos A} + \frac{\sin B}{\cos B} = \frac{\sin A \cdot \cos B + \cos A \sin B}{\cos A \cdot \cos B} \\ &= \frac{\sin(A+B)}{\cos(90-B)\cos(90-A)} \\ &= \frac{1}{\sin A \cdot \sin B} = \frac{4R^2}{ab} \\ \frac{C}{\sin 90} &= 2R \text{ (sin Rule)} \\ R &= \frac{C}{2} \\ &= \frac{4\left(\frac{C}{2}\right)^2}{ab} = \frac{a}{bc}\end{aligned}$$

Q2. In a triangle ABC if $\frac{\cos A}{a} + \frac{\cos B}{b} + \frac{\cos C}{c} = \frac{a}{bc}$, then A is

- (A) an acute angle (B) an obtuse angle (C) a right angle (D) equal to B - C

Ans. C

Sol. $\frac{\cos A}{a} + \frac{\cos B}{b} + \frac{\cos C}{c} = \frac{a}{bc}$

$$\frac{b^2 + c^2 - a^2}{2abc} + \frac{a^2 + c^2 - b^2}{2abc} + \frac{a^2 + b^2 - c^2}{2abc} = \frac{a}{bc}$$

$$a^2 + b^2 + c^2 = 2a^2$$

$$a^2 = b^2 + c^2$$

$$\angle = 90^\circ$$

Q3. If in a ΔABC , $\sin^3 A + \sin^3 B + \sin^3 C = 3 \sin A \cdot \sin B \cdot \sin C$, then

- (A) ΔABC may be a scalene triangle (B) ΔABC is a right triangle
(C) ΔABC is an obtuse angled triangle (D) ΔABC is an equilateral triangle

Ans. D

Sol. Use : $x^3 + y^3 + z^3 - 3xyz = (1/2)(x+y+z)[(x-y)^2 + (y-z)^2 + (z-x)^2]$ where $x = \sin A$ etc.

either $\sin A + \sin B + \sin C = 0$ (which is not possible)
or $\sin A = \sin B = \sin C \Rightarrow$ equilateral

Q4. In a triangle ABC, $a : b : c = 4 : 5 : 6$. Then $3A + B =$

- (A) $4C$ (B) 2π (C) $\pi - C$ (D) π

Ans. D

Sol. $a = 4k, b = 5k, c = 6k$

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{25k^2 + 36k^2 - 16k^2}{60k^2} = \frac{45}{60} = \frac{3}{4}$$

$$\cos 2A = 2\cos^2 A - 1 = 2 \cdot \frac{9}{16} - 1 = \frac{1}{8}$$

$$\cos c = \frac{a^2 + b^2 - c^2}{2ab} = \frac{16k^2 + 25k^2 - 36k^2}{40k^2} = \frac{1}{8}$$

$$\cos c = \cos 2A$$

$$\angle C = \angle 2A$$

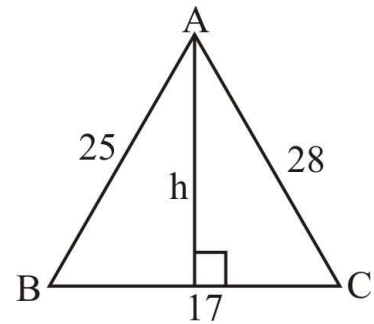
$$3A + B = A + 2A + B = A + B + C = \pi$$

Q5. The sides of a triangle are 17, 25 and 28. The length of the largest altitude is :

- (A) 15 (B) $84/5$ (C) $420/17$ (D) $210/17$

Ans. C

Sol.



Largest altitude to the smallest side

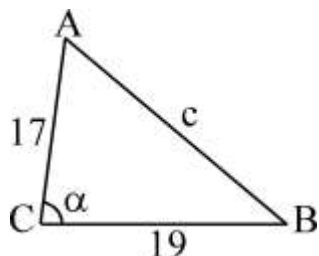
$$S = \frac{17 + 28 + 25}{2} = 35$$

$$\Delta = \frac{1}{2} 17 \times h = \sqrt{5(5-a)(5-b)(5-c)}$$

$$\frac{17h}{h} = \sqrt{35 \times 18 \times 7 \times 10}$$

$$h = \frac{420}{17}$$

Q6. Consider the triangle pictured as shown. If $0 < \alpha < \pi/2$ then the number of integral values of c is



- (A) 35 (B) 23 (C) 24 (D) 25

Ans. B

Sol. $\cos C = \frac{(19)^2 + (17)^2 - c^2}{2 \cdot 19 \cdot 17}$

$$\begin{aligned} \therefore c^2 &< (19)^2 + (17)^2 \\ c^2 &< (18+1)^2 + (18-1)^2 \\ c^2 &< 2(18^2 + 1) \\ c^2 &< 650 \quad \dots(1) \end{aligned}$$

$$c > 2 \quad \text{and} \quad c < 26$$

$$\therefore 2 < c < 26 \quad \dots(2)$$

$$4 < c^2 < 676 \quad \dots(3)$$

$$\therefore c \text{ is } 3, 4, 5, \dots, 25$$

$$\therefore \text{number of integral values of } c \text{ is } 23$$

Q7. Triangle ABC has $BC = 1$ and $AC = 2$. The maximum possible value of the angle A is

- (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$

Ans. A

Sol. -

Q8. Let ABC be a triangle right angled at C. The value of $\frac{\log_{b+c} a + \log_{c-b} a}{\log_{b+c} a \cdot \log_{c-b} a}$ ($b + c \neq 1, c - b \neq 1$) equals

- (A) 1 (B) 2 (C) 3 (D) 1/2

Ans. B

Sol. $\angle C = 90^\circ$

$$a^2 + b^2 = c^2$$

$$= \frac{\log_{b+c}^a + \log_{c-b}^a}{\log_{b+c}^a \cdot \log_{c-b}^a}$$

$$= \frac{1}{\log_{c-b}^a} + \frac{1}{\log_{b+c}^a}$$

$$= \log_a (c-b) + \log_a (b+c)$$

$$= \log_a (c-b)(c+b) = \log_a (c^2 - b^2) = \log_a a^2 = 2$$

Q9. In a triangle ABC, angle B < angle C and the values of B and C satisfy the equation $2 \tan x - k(1 + \tan^2 x) = 0$ where $(0 < k < 1)$. Then the measure of angle A is :

- (A) $\pi/3$ (B) $2\pi/3$ (C) $\pi/2$ (D) $3\pi/4$

Ans. C

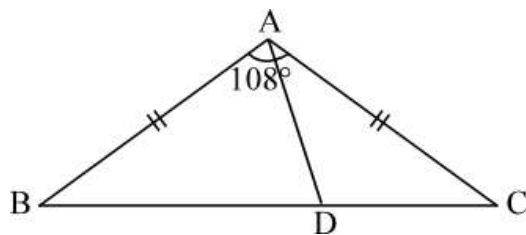
Sol. $k = \frac{2 \tan x}{1 + \tan^2 x} = \sin 2x \Rightarrow \sin 2C = \sin 2B$

But $\angle C > \angle B$

$$2C = \pi - 2B \Rightarrow B + C = \pi/2$$

$$\therefore \angle A = \pi/2$$

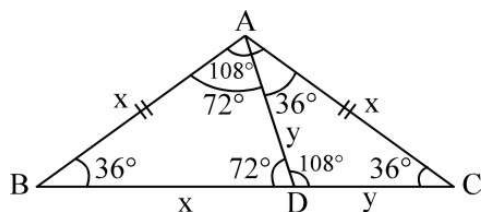
Q10. In an isosceles triangle ABC, $AB = AC$, $\angle BAC = 108^\circ$ and AD trisects $\angle BAC$ and $BD > DC$. The ratio $\frac{BD}{DC}$ is



- (A) $\frac{3}{2}$ (B) $\frac{\sqrt{5} + 1}{2}$ (C) $\sqrt{5} - 1$ (D) 2

Ans. B

Sol.



In $\triangle ABC \left(\frac{x}{y} = ? \right)$

$$\frac{x+y}{\sin 108^\circ} = \frac{x}{\sin 36^\circ}$$

$$\frac{x+y}{x} = \frac{\sin 108^\circ}{\sin 36^\circ} \dots\dots\dots (1)$$

$$1 + \frac{y}{x} = \frac{\sin 72^\circ}{\sin 36^\circ} = 2 \cos 36^\circ$$

$$\frac{x}{y} = \frac{2(\sqrt{5}+1)}{(\sqrt{5}-1)(\sqrt{5}+1)} = \frac{\sqrt{5}+1}{2}$$

Q11. The ratio of the sides of a triangle ABC is $1 : \sqrt{3} : 2$. The ratio $A : B : C$ is

- (A) $3 : 5 : 2$ (B) $1 : \sqrt{3} : 2$ (C) $3 : 2 : 1$ (D) $1 : 2 : 3$

Ans. D

Sol. $\sin A : \sin B : \sin C = a : b : c$

$$= 1 : \sqrt{3} : 2$$

$$\sin A : \sin B : \sin C = \frac{1}{2} : \frac{\sqrt{3}}{2} : 1$$

$$\sin A = \frac{1}{2} \quad \sin B = \frac{\sqrt{3}}{2}, \quad \sin C = 1$$

$$A = 30 \quad A = 60 \quad C = 90$$

$$A : B : C = 30 : 60 : 90$$

$$= 1 : 2 : 3$$

Q12. In $\triangle ABC$, a, b, c are the lengths of its sides and A, B, C are the angles of triangle ABC. The correct relation is

- (A) $(b-c)\sin\left(\frac{B-C}{2}\right) = a \cos\left(\frac{A}{2}\right)$ (B) $(b-c)\cos\left(\frac{A}{2}\right) = a \sin\left(\frac{B-C}{2}\right)$
 (C) $(b+c)\sin\left(\frac{B+C}{2}\right) = a \cos\left(\frac{A}{2}\right)$ (D) $(b-c)\cos\left(\frac{A}{2}\right) = 2a \sin\left(\frac{B+C}{2}\right)$

Ans. B

Sol. (A) $(b-c) \cdot \sin\left(\frac{B-C}{2}\right)$

$$2R (\sin B - \sin C) \sin\left(\frac{B-C}{2}\right)$$

$$2R 2\cos\left(\frac{B+C}{2}\right) \cdot \sin\left(\frac{B-C}{2}\right) \cdot \sin\left(\frac{B-C}{2}\right)$$

$$4R \sin \frac{A}{2} \cdot \sin^2\left(\frac{B-C}{2}\right) \neq \text{RHS}$$

$$\begin{aligned}
& \text{(B)} \quad (b - c) \cos \frac{A}{2} \\
& = 4R \sin \frac{A}{2} \cdot \sin \left(\frac{B-C}{2} \right) \cdot \cos \frac{A}{2} \\
& = 2R \sin A \cdot \sin \left(\frac{B-C}{2} \right) \\
& = a \cdot \sin \left(\frac{B-C}{2} \right)
\end{aligned}$$

Q13. If in a ΔABC , $\cos A \cdot \cos B + \sin A \sin B \sin 2C = 1$ then, the statement which is incorrect, is

- (A) ΔABC is isosceles but not right angled (B) ΔABC is acute angled
 (C) ΔABC is right angled (D) least angle of the triangle is $\pi/4$

Ans. C

Sol. $\sin 2C = \frac{1 - \cos A \cos B}{\sin A \sin B} \leq 1$. Now proceed $A = B = \frac{3\pi}{8}$ and $C = \frac{\pi}{4}$
 $1 \leq \cos \Rightarrow (A - B) \cos (A - B) = 1 \Rightarrow A = B$

Q14. The product of the arithmetic mean of the lengths of the sides of a triangle and harmonic mean of the lengths of the altitudes of the triangle is equal to :

- (A) Δ (B) 2Δ (C) 3Δ (D) 4Δ

Ans. B

Sol. where Δ is the area of the triangle ABC

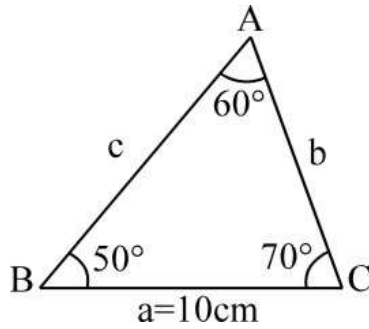
$$ah_1 = bh_2 = ch_3 = 2\Delta \Rightarrow \frac{1}{h_1} + \frac{1}{h_2} + \frac{1}{h_3} = \frac{a+b+c}{3} \Rightarrow \frac{3}{\frac{1}{h_1} + \frac{1}{h_2} + \frac{1}{h_3}} = 2\Delta$$

Q15. A triangle has base 10 cm long and the base angles of 50° and 70° . If the perimeter of the triangle is $x + y \cos z^\circ$ where $z \in (0, 90)$ then the value of $x + y + z$ equals

- (A) 60 (B) 55 (C) 50 (D) 40

Ans. D

Sol.



Given $P = x + y \cos z^\circ$

Also $P = 10 + b + c \dots(1)$

Now using sine law

$$\frac{10}{\sin 60^\circ} = \frac{b}{\sin 50^\circ} = \frac{c}{\sin 70^\circ}$$

$$\therefore b = \frac{10 \sin 50^\circ}{\sin 60^\circ} \text{ and } c = \frac{10 \sin 70^\circ}{\sin 60^\circ}$$

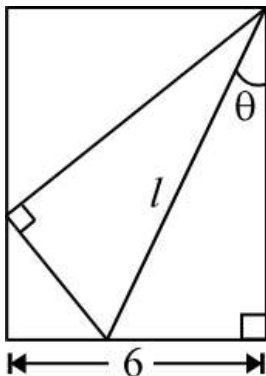
$$\begin{aligned} b + c &= \frac{10}{\sin 60^\circ} (\sin 50^\circ + \sin 70^\circ) \\ &= \frac{10}{\sin 60^\circ} [2 \sin 60^\circ \cos 10^\circ] = 20 \cos 10^\circ \end{aligned}$$

$$\therefore P = 10 + 20 \cos 10^\circ$$

$$\therefore x = 10; \quad y = 20; \quad z = 10$$

$$x + y + z = 40$$

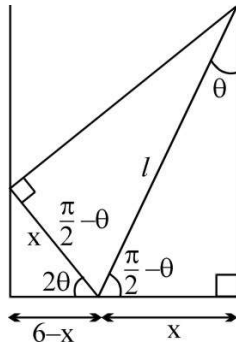
- Q16. One side of a rectangular piece of paper is 6 cm, the adjacent sides being longer than 6 cms. One corner of the paper is folded so that it sets on the opposite longer side. If the length of the crease is 1 cms and it makes an angle θ with the long side as shown, then l is



- (A) $\frac{3}{\sin \theta \cos^2 \theta}$ (B) $\frac{6}{\sin \theta \cos^2 \theta}$ (C) $\frac{3}{\sin \theta \cos \theta}$ (D) $\frac{3}{\sin^3 \theta}$

Ans. A

Sol.



$$\sin \theta = \frac{x}{l} \quad \dots\dots (1); \quad \text{also } \cos 2\theta = \frac{6-x}{x}$$

$$1 + \cos 2\theta = \frac{6}{x};$$

$$2 \cos^2 \theta = \frac{6}{l \sin \theta} \quad \{\text{substituting } x = l \sin \theta \text{ from (1)}\}$$

$$l = \frac{3}{\sin \theta \cos^2 \theta}$$

Q17. Angles A, B and C of a triangle ABC are in A.P. If $\frac{b}{c} = \sqrt{\frac{3}{2}}$, then $\angle A$ is equal to

- (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{5\pi}{12}$ (D) $\frac{\pi}{2}$

Ans. C

Sol. A, B, C in A

$$2B = A + C$$

$$\angle B = 60$$

$$\frac{\sin B}{\sin C} = \frac{b}{c}$$

$$\frac{\sin 60}{\sin C} = \sqrt{\frac{3}{2}}$$

$$\sin C = \frac{\sqrt{3}}{2} \cdot \sqrt{\frac{2}{3}} = \frac{1}{\sqrt{2}}$$

$$C = \frac{\pi}{4} = 45^\circ \quad \left(\frac{3\pi}{4}\right) \text{ Not possible}$$

$$A = 75^\circ$$

Q18. In a triangle ABC, $\angle A = 60^\circ$ and $b : c = (\sqrt{3} + 1) : 2$ then $(\angle B - \angle C)$ has the value equal to

- (A) 15° (B) 30° (C) 22.5° (D) 45°

Ans. B

Sol. -

- Q19. In a triangle ABC, if $A = 30^\circ$ and $\frac{b}{c} = \frac{2 + \sqrt{3} + \sqrt{2} - 1}{2 + \sqrt{3} - \sqrt{2} + 1}$, then C, is equal to
- (A) 75° (B) 15° (C) $52\frac{1^\circ}{2}$ (D) $97\frac{1^\circ}{2}$

Ans. C

Sol. $A = 30$

$$B + C = 150$$

$$\begin{aligned}\tan\left(\frac{B-C}{2}\right) &= \frac{b-c}{b+c} \cot\left(\frac{A}{2}\right) \\ &= \frac{\frac{b}{c} - 1}{\frac{b}{c} + 1} \cot\left(\frac{A}{2}\right) \\ &= \frac{2(\sqrt{2}-1)}{2(\sqrt{3}+1)} \cdot \cot(15) \\ &= \left(\frac{\sqrt{2}-1}{2+\sqrt{3}}\right) \left(\frac{1+\cos 30}{\sin 30}\right) = \left(\frac{\sqrt{2}-1}{2+\sqrt{3}}\right) (2+\sqrt{3})\end{aligned}$$

$$\tan\left(\frac{B-C}{2}\right) = (\sqrt{2}-1)$$

$$\left(\frac{B-C}{2}\right) = \frac{\pi}{8}$$

$$B - C = 45^\circ$$

$$2C = 150^\circ$$

$$C = 52\frac{1^\circ}{2}$$

- Q20. If the angles A, B and C of a triangle are in an arithmetic progression and if a, b and c denote the lengths of the sides opposite to A, B and C respectively, then the value of the expression $\frac{a}{c} \sin 2C + \frac{c}{a} \sin 2A$, is

- (A) $\frac{1}{2}$ (B) $\frac{\sqrt{3}}{2}$ (C) 1 (D) $\sqrt{3}$

Ans. D

Sol. $B = 60^\circ$

$$\begin{aligned}\therefore \frac{a}{c} \sin 2C + \frac{c}{a} \sin 2A &= 2 \sin A \cos C + 2 \sin C \cos A \\ &= 2 \sin (A + C) = 2 \sin B = 2 \times \frac{\sqrt{3}}{2} = \sqrt{3}.\end{aligned}$$

- Q21. Let ABC be a triangle such that $\angle ACB = \frac{\pi}{6}$ and let a, b and c denote the lengths of the sides opposite to A, B and C respectively. The value(s) of x for which $a = x^2 + x + 1$, $b = x^2 - 1$ and $c = 2x + 1$ is/are

- (A) $-(2 + \sqrt{3})$ (B) $1 + \sqrt{3}$ (C) $2 + \sqrt{3}$ (D) $4\sqrt{3}$

Ans. B

Sol. $\cos C = \frac{a^2 + b^2 - c^2}{2ab}$

$$\frac{\sqrt{3}}{2} = \frac{(x^2 + x + 1)^2 + (x^2 - 1)^2 - (2x + 2)^2}{2(x^2 + x + 1)(x^2 - 1)}$$

$$\Rightarrow x^4(2 - \sqrt{3}) + x^3(2 - \sqrt{3}) - 3x^2 - x(2 - \sqrt{3}) + (\sqrt{3} + 1) = 0$$

$$\Rightarrow (x^2 - 1)[(2 - \sqrt{3})x^2 + (2 - \sqrt{3})x - (\sqrt{3} + 1)] = 0$$

Now

$$\begin{cases} x^2 + x + 1 + x^2 - 1 > 2x + 1 \\ x^2 + x + 1 + 2x + 1 > x^2 - 1 \\ 2x + 1 + x^2 - 1 > x^2 + x + 1 \end{cases}$$

($\therefore$ sum of two sides is greater than third side)

$$\Rightarrow x > 1 \quad \Rightarrow x = 1 + \sqrt{3}$$

Alternate :

$$\tan = \frac{\pi}{12} \sqrt{\frac{(s-b)(s-a)}{s(s-c)}}$$

$$s = x^2 + \frac{3x+1}{2}$$

$$s - b = \frac{3(x+1)}{2}, s - a = \frac{x-1}{2}, s - c = x^2 - \frac{(x+1)}{2}$$

$$\tan \frac{\pi}{12} = \sqrt{\frac{\frac{3}{2}(x+1)\frac{(x-1)}{2}}{(x^2 + \frac{3x+1}{2})(x^2 - \frac{x+1}{2})}}$$

Simplifying

$$2 - \sqrt{3} = \frac{\sqrt{3}}{2x+1}$$

$$x = \sqrt{3} + 1$$

Q22. The expression $\frac{(a+b+c)(c+a-b)(a+b-c)}{4b^2c^2}$ is equal to

- (A) $\cos 2A$ (B) $1 + \cos A$ (C) $1 - \cos A$ (D) $\sin^2 A$

Ans. D

Sol. $= \frac{(a+b+c)(b+c-a)(a+b-c)}{4b^2c^2}$

$$\begin{aligned}
&= \frac{(2s)(2s-2a)(2s-2b)(2s-2c)}{4b^2c^2} \\
&= \frac{4s(s-a)(s-b)(s-c)}{b^2c^2} \\
&= \frac{4\Delta^2}{b^2c^2} \\
&= \frac{4 \cdot \frac{1}{4} \cdot b^2c^2 \sin^2 A}{b^2c^2} \\
&= \sin^2 A
\end{aligned}$$

Q23. In a ΔABC , if $\frac{s-a}{11} = \frac{s-b}{12} = \frac{s-c}{13}$, then $\tan^2 \frac{A}{2}$ is equal to

- (A) $\frac{143}{342}$ (B) $\frac{13}{33}$ (C) $\frac{11}{99}$ (D) $\frac{12}{37}$

Ans. B

Sol. Let $\frac{s-a}{11} = \frac{s-b}{12} = \frac{s-c}{13} = k$

$$S = 36k$$

$$\tan \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}} = \sqrt{\frac{(12k)(13k)}{36k(11k)}}$$

$$\tan^2 \frac{A}{2} = \frac{13}{33}$$

Q24. If a, b, c are in A.P. then the numerical value of $\tan \frac{A}{2} \tan \frac{C}{2}$ is :

- (A) $1/4$ (B) $1/3$ (C) $1/2$ (D) 1

Ans. B

Sol. $2b = a + c$ (a, b, c in AP)

$$2s = 3b$$

$$\begin{aligned}
\tan \frac{A}{2} \cdot \tan \frac{C}{2} &= \sqrt{\frac{(s-b)(s-c)}{s(s-a)}} \times \frac{(s-a)(s-b)}{s(s-c)} = \frac{s-b}{s} \\
&= \frac{s - \frac{2}{3}s}{s} = \frac{1}{3}
\end{aligned}$$

Q25. In ΔABC , the minimum value of $\frac{\sum \cot^2 \frac{A}{2} \cdot \cot^2 \frac{B}{2}}{\prod \cot^2 \frac{A}{2}}$ is

- (A) 1 (B) 2 (C) 3 (D) non existent

Ans. A

Sol. $E = \tan^2 \frac{A}{2} + \tan^2 \frac{B}{2} + \tan^2 \frac{C}{2}$

now consider $\left(\tan \frac{A}{2} - \tan \frac{B}{2}\right)^2 \geq 0$ etc and add to get the result.

Q26. In a triangle ABC, right angles at B, the inradius is :

- (A) $\frac{AB + BC - AC}{2}$ (B) $\frac{AB + AC - CC}{2}$ (C) $\frac{AB + BC + AC}{2}$ (D) $\frac{BC + AC - AB}{2}$

Ans. A

Sol. $\angle B = 90^\circ$

$$r = (s - b) \tan \frac{B}{2}$$

$$r = (s - b) \tan 45^\circ$$

$$r = \frac{2s - 2b}{2} = \frac{a + c - c}{2} = \frac{CB + AB - AC}{2}$$

Q27. In a ΔABC , $\frac{r_1 + r_2}{1 + \cos C}$ is equal to

- (A) $\frac{abc}{2\Delta}$ (B) $2R$ (C) $\frac{abc}{\Delta^2}$ (D) $\frac{abc}{\Delta}$

Ans. A

$$\begin{aligned} \text{Sol. } \frac{r_1 + r_2}{1 + \cos C} &= \frac{\frac{\Delta}{s-a} + \frac{\Delta}{s-b}}{2\cos^2 \frac{A}{2}} \\ &= \frac{\Delta(s-a+s-b)}{2 \cdot (s-a)(s-b) \frac{s(s-a)}{bc}} \\ &= \frac{\Delta abc}{2\Delta^2} = \frac{abc}{2\Delta} \end{aligned}$$

Q28. In a ΔABC , $r_1 r_2 + r r_3 =$

- (A) $\frac{ab}{c}$ (B) abc (C) ab (D) $\frac{(a+b)}{c}$

Ans. C

$$\begin{aligned} \text{Sol. } r_1 r_2 + r r_3 &= \frac{\Delta}{(s-a)} \times \frac{\Delta}{(s-b)} + \frac{\Delta \cdot \Delta}{s(s-c)} \\ &= \frac{\Delta^2(s^2 - sc + s^2 - sa - sb + ab)}{s(s-a)(s-b)(s-c)} \\ &= \frac{\Delta^2(2s^2 - s(a+b+c) - ab)}{\Delta^2} \\ &= ab \end{aligned}$$

Q29. In a triangle ABC, $R(b+c) = a\sqrt{bc}$ where R is the circumradius of the triangle. Then the triangle is

- (A) Isosceles but not right (B) right but not isosceles (C) right isosceles

(D) equilateral

Ans. C

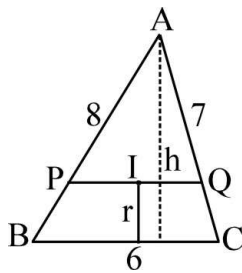
Sol. $R(b + c) = a \sqrt{bc}$
 $R(b + c) = 2R \sin A \sqrt{bc}$
 $\therefore \sin A = \frac{b + c}{2\sqrt{bc}}$
 now applying AM $\geq$ GM for b and c
 $\frac{b + c}{2} \geq \sqrt{bc} \Rightarrow b + c \geq 2\sqrt{bc}$
 hence $\sin A \geq 1$ which is not possible.
 hence $\sin A = 1 \Rightarrow A = 90^\circ$
 $\therefore A = 90^\circ$ and $b = c \Rightarrow (C)$

Q30. A triangle has sides 6, 7, 8. The line through its incentre parallel to the shortest side is drawn to meet the other two sides at P and Q. The length of the segment PQ is

- (A) $\frac{12}{5}$ (B) $\frac{15}{4}$ (C) $\frac{30}{7}$ (D) $\frac{33}{9}$

Ans. C

Sol.



$$\Delta = \frac{6 \cdot h}{2} = 3h ; (\text{where } h \text{ is the altitude from } A)$$

$$\text{Also } \Delta = \frac{21 \cdot r}{2} \text{ (using } \Delta = r \cdot s)$$

$$\therefore \frac{21 \cdot r}{2} = \frac{6 \cdot h}{2} = 3h \Rightarrow \frac{r}{h} = \frac{2}{7}$$

now APQ and ABC are similar

$$\frac{h - r}{h} = \frac{PQ}{6} \Rightarrow 1 - \frac{PQ}{6} \Rightarrow 1 - \frac{2}{7} = \frac{PQ}{6} \Rightarrow \frac{5}{7} \Rightarrow PQ = \frac{30}{7}$$

Q31. If r_1, r_2, r_3 be the radii of excircles of the triangle ABC, then $\frac{\sum r_1}{\sqrt{\sum r_1 r_2}}$ is equal to :

- (A) $\sum \cot \frac{A}{2}$ (B) $\sum \cot \frac{A}{2} \cot \frac{B}{2}$ (C) $\sum \tan \frac{A}{2}$ (D) $\prod \tan \frac{A}{2}$

Ans. C

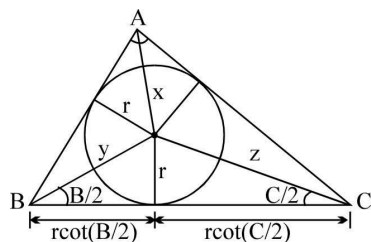
Sol. $\frac{s \sum \tan \frac{A}{2}}{s^2} = \sum \tan \frac{A}{2} \Rightarrow C$

Q32. If x, y & z are the distances of incentre from the vertices of the ΔABC respectively then $\frac{abc}{xyz} =$

(A) $\prod \tan \frac{A}{2}$ (B) $\sum \cot \frac{A}{2}$ (C) $\sum \tan \frac{A}{2}$ (D) $\sum \sin \frac{A}{2}$

Ans. B

Sol.



$$x = r \operatorname{cosec} \frac{A}{2}$$

$$a = r \left(\cot \frac{B}{2} + \cot \frac{C}{2} \right) \cdot \sin \frac{A}{2} = \frac{\sin \frac{A}{2} \cdot \cos \frac{A}{2}}{\sin \frac{B}{2} \cdot \sin \frac{C}{2}}$$

$$\therefore \frac{abc}{xyz} = \frac{\cos \frac{A}{2} \cdot \cos \frac{B}{2} \cdot \cos \frac{C}{2}}{\sin \frac{A}{2} \sin \frac{B}{2} \cdot \sin \frac{C}{2}} = \cot \frac{A}{2} \cdot \cot \frac{B}{2} \cdot \cot \frac{C}{2}$$

$$\text{In a triangle} = \prod \tan \frac{A}{2} \sum \tan \frac{A}{2}$$

Q33. With usual notations, in a triangle ABC, $a \cos(B - C) + b \cos(C - A) + c \cos(A - B)$ is equal to

(A) $\frac{abc}{R^2}$ (B) $\frac{abc}{4R^2}$ (C) $\frac{4abc}{R^2}$ (D) $\frac{abc}{2R^2}$

Ans. A

Sol. Put $a = 2R \sin A$ etc.

$$\begin{aligned} E &= R[\sin 2B + \sin 2C + \sin 2C + \sin 2A + \sin 2A + \sin 2B] \\ &= 2R(\sin 2A + \sin 2B + \sin 2C) \\ &= 8R \cdot \frac{a}{2R} \cdot \frac{b}{2R} \cdot \frac{c}{2R} = \frac{abc}{R^2} \end{aligned}$$

Alternatively: $a = 2R \sin A = 2R \sin(B + C)$

$$R[(2 \sin(B + C) \cos(B - C) + \dots + \dots)]$$

$$R[(\sin 2B + \sin 2C) + \dots + \dots]$$

$$2R(\sin 2A + \sin 2B + \sin 2C)$$

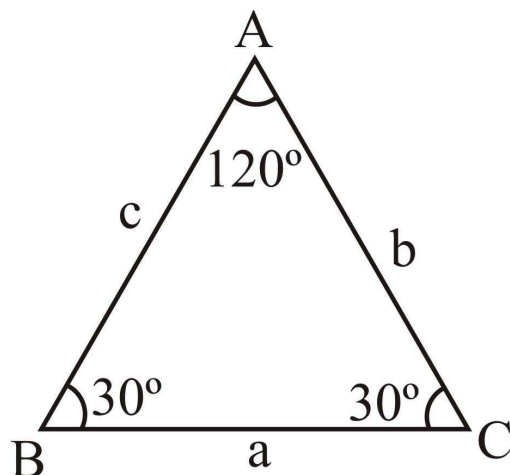
$$8R \sin A \cdot \sin B \cdot \sin C = \frac{abc}{R^2}$$

Q34. Given an isosceles triangle, whose one angle is 120° and radius of its incircle is $\sqrt{3}$. Then the area of triangle in sq. units is

- (A) $7 + 12\sqrt{3}$ (B) $12 - 7\sqrt{3}$ (C) $12 + 7\sqrt{3}$ (D) 4π

Ans. C

Sol.



$$r = (s - a) \tan \frac{A}{2}$$

$$\sqrt{3} = (s - a) \tan 60$$

$$s - a = 1 \dots\dots\dots(1)$$

$$s - b = \frac{\sqrt{3}}{\tan 15^\circ} = \frac{\sqrt{3}}{2 - \sqrt{3}} = \frac{\sqrt{3}(2 + \sqrt{3})}{1} = 2\sqrt{3} + 3 \dots\dots\dots(2)$$

$$s - c = 2\sqrt{3} + 3 \dots\dots\dots(3)$$

$$(1) + (2) + (3)$$

$$S = 4\sqrt{3} + 7$$

$$b = cs - (2\sqrt{3} + 3)$$

$$= 4\sqrt{3} + 4 = 2(\sqrt{3} + 2)$$

$$\Delta = \frac{1}{2}bc \sin A$$

$$= \frac{1}{2} 4(\sqrt{3} + 2)^2 \frac{\sqrt{3}}{2}$$

$$= \sqrt{3}(3 + 4 + 4\sqrt{3})$$

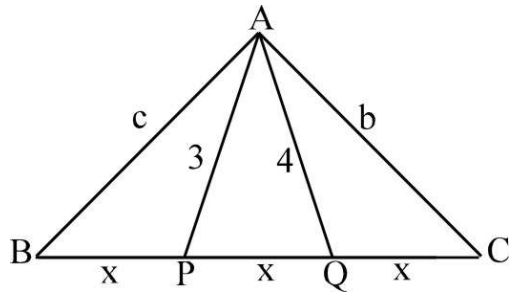
$$= 7\sqrt{3} + 12$$

Q35. Triangle ABC is right angled at A. The points P and Q are on the hypotenuse BC such that BP = PQ = QC. If AP = 3 and AQ = 4 then the length BC is equal to

- (A) $\sqrt{27}$ (B) $\sqrt{36}$ (C) $\sqrt{45}$ (D) $\sqrt{54}$

Ans. C

Sol.



In $\triangle ABP$

$$9 = c^2 + x^2 + -2cx \cos B; \quad \text{but } \cos B = \frac{c}{3x}$$

$$9 = c^2 + x^2 + -2cx \frac{c}{3x}$$

$$9 = x^2 + \frac{c^2}{3} \dots\dots\dots(1); \quad |||ly \quad 16x = x^2 + \frac{b^2}{3} \dots\dots\dots(2)$$

$$(1) + (2)$$

$$25 = 2x^2 + (b^2 + c^2) = 2x^2 + 3x^2$$

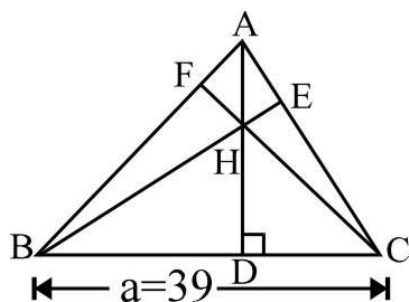
$$\Rightarrow 5x^2 = 25 \quad \Rightarrow x = \sqrt{5}; \quad \therefore BC = 3\sqrt{5} = \sqrt{45}$$

- Q36. In an acute angled triangle ABC, point D, E and F are the feet of the perpendiculars from A, B and C onto BC, AC and AB respectively. H is the intersection of AD and BE. If $\sin A = 3/5$ and $BC = 39$, the length of AH is

- (A) 45 (B) 48 (C) 52 (D) 54

Ans. C

Sol.



$$AH = 2R \cos A \quad (\text{distance of orthocentre from the vertex})$$

$$\text{given } \sin A = 3/5; \quad \cos A = 4/5$$

$$a = 39$$

$$\therefore \frac{a}{\sin A} = 2R; = \frac{39.5}{3} = 2R \quad \Rightarrow R = 65$$

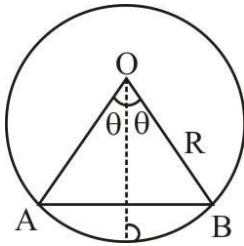
$$\therefore AH = 65. \frac{4}{5} = 52$$

- Q37. If R is the radius of circumscribing circle of a regular polygon of n-sides with side length 'a' then R =

- (A) $\frac{a}{2} \sin\left(\frac{\pi}{n}\right)$ (B) $\frac{a}{2} \sin\left(\frac{\pi}{2n}\right)$ (C) $\frac{a}{2} \cos \sec\left(\frac{\pi}{n}\right)$ (D) $\frac{a}{2} \cos \sec\left(\frac{\pi}{2n}\right)$

Ans. C

Sol.



let side length AB = a

$$2Q = \frac{2\pi}{n}$$

$$Q = \frac{\pi}{n}$$

$$PB = \frac{a}{2}$$

$$\sin \theta = \frac{PB}{R}$$

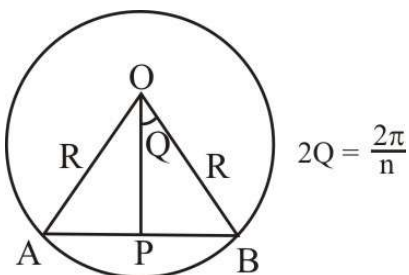
$$R = \frac{a/2}{\sin \frac{\pi}{n}} = \frac{a}{2} \operatorname{cosec} \frac{\pi}{n}$$

Q38. If A_1, A_2, A_3 denote respectively the areas of an inscribed polygon of $2n$ sides, inscribed polygon of n sides and circumscribed polygon of n sides, then A_2, A_1, A_3 are in

- (A) AP (B) GP (C) HP (D) none of these

Ans. B

Sol.



Let radius of circle is R

for A_2 ,

$$\Delta AOB = \frac{1}{2} R^2 \sin 2\theta$$

$$A_2 = n \cdot \frac{nR^2}{2} = \sin\left(\frac{2\pi}{n}\right)$$

for A_1 put $n = 2n$

$$A_1 = nR^2 \sin \left(\frac{\pi}{n} \right)$$

for A_3 $2Q = \frac{27}{n}$

$$\tan \theta = \frac{PB}{R}$$

$$PB = R \tan \theta$$

$$AB = 2R \tan \frac{\pi}{n}$$

$$A_3 = n \left(\frac{1}{2} 2R \tan \frac{\pi}{n} \cdot R \right)$$

$$= nR^2 \tan \frac{\pi}{n}$$

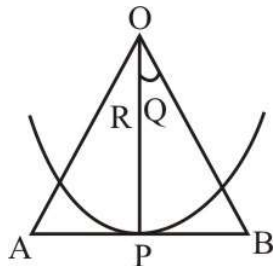
$$A_2 \cdot A_3 = \left(na^2 \sin \frac{\pi}{n} \right)^2$$

Q39. The area of the circle and the area of a regular polygon of n sides and of perimeter equal to that of the circle are in the ratio of

(A) $\tan \left(\frac{\pi}{n} \right) : \frac{\pi}{n}$ (B) $\cos \left(\frac{\pi}{n} \right) : \frac{\pi}{n}$ (C) $\sin \frac{\pi}{n} : \frac{\pi}{n}$ (D) $\cot \left(\frac{\pi}{n} \right) : \frac{\pi}{n}$

Ans. A

Sol.



Let side of polygon = a

Let radius of circle = R

$$2\pi R = na$$

$$\text{Area of circle} = \pi R^2$$

Area of inscribed polygon of n sides

$$= \frac{na^2}{4} \cot \frac{\pi}{n}$$

$$\frac{\pi R^2}{\frac{na^2}{4} \cot \frac{\pi}{n}} = \frac{\frac{\pi n^2 a^2}{4\pi^2}}{n \frac{a^2}{4} \cot \frac{\pi}{n}}$$

$$= \frac{\tan(\pi/n)}{\left(\frac{\pi}{n} \right)}$$